SOFTWARE ENGINEERING MINI-PROJECT

Deliverables Part 1

CODECRAFT

EARLY METABOLIC RISK DETECTION WEBSITE

ML vs DL vs QML comparison on NHANES 2017-2018

Part A: Software Requirements Specification · Part B: Software Architecture and Design · Part C: Software Test Plan and Test Cases

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Part A – Software Requirements Specification (SRS)

Prepared in the style of IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018. Product: **Early Metabolic Risk Detection Website** that builds and compares ML, DL and QML models.

1. Introduction

1.1 Purpose

This SRS states what the Early Metabolic Risk Detection website shall do. It is written for the CodeCraft team, the course instructor and reviewers. The test plan in Part C verifies every requirement written here.

1.2 Scope

The product is a browser-based academic website. An authorized student or research user can load a prepared, de-identified dataset, validate it, train and evaluate three separate classifiers (classical ML, deep learning and quantum ML), and compare their results side by side on the same held-out records. The website may give an optional elevated-risk estimate for one record. It **does not** diagnose disease, recommend treatment, replace a clinician or claim clinical readiness.

Project goal: show an honest ML vs DL vs QML comparison. This document does not assume which model will win; results are added only after experiments are run.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Metabolic risk	Project label: elevated-risk group when HbA1c (LBXGH) is in the prediabetes range 5.7% to <6.5%; normal group when LBXGH < 5.7%.
ML / DL / QML	Classical machine learning (e.g. logistic regression, random forest); deep learning (feed-forward neural network); quantum machine learning (variational quantum classifier run on a simulator).
NHANES	U.S. National Health and Nutrition Examination Survey, 2017-2018 public-use cycle.
ROC-AUC	Area under the receiver operating characteristic curve.
Sensitivity / Specificity	True-positive rate / true-negative rate for the elevated-risk class.
Leakage	Use of information during training that would not be available at prediction time.
Held-out test set	Records never used for fitting, feature selection, threshold selection or tuning.
SRS / STP / SAD	Software Requirements Specification / Software Test Plan / Software Architecture and Design Specification.

1.4 Overview

Section 2 describes the product context. Section 3 gives the specific requirements: interfaces, use cases, functional and non-functional requirements, performance, constraints and the security section. Section 4 records the dataset and website details. Part B is the architecture and design specification; Part C is the test plan with test cases and traceability.

2. Overall Description

2.1 Product perspective

The product is a standalone website run locally for the course demonstration. Workflow: load the prepared NHANES data, validate it, run the separate ML, DL and QML pipelines, and compare their results on the same test records. The three models stay independent; the data split and scoring rules are shared.

Interface type	Description
System	Modern desktop browser (latest Chrome or Firefox) talking to a local application server. Provides dataset information, validation, model runs, a comparison dashboard and report export.
User	Landing page, data validation summary, controls to run one or all models, comparison view with labelled metrics and plots. Every individual estimate carries a plain-language non-diagnostic warning. No personal identity details are requested.
Hardware	General-purpose laptop or desktop. GPU optional for DL. QML runs on a simulator by default; a real quantum processor is optional and used only if accessible and approved. Hardware, backend and runtime are recorded for each run.
Software	Documented language and library versions for data processing, ML, DL and QML. Input is a documented CSV schema. No clinical-system integration.
Communications	Core processing is local and needs no network. If any web service is used, it shall use HTTPS, restricted access, and no third-party transfer of health records without project approval.
Memory	Peak memory is recorded per run. If resources are exceeded the run stops gracefully, reports the limit and leaves earlier results intact.
Operations	User opens the site, checks the dataset summary, chooses one model or all three, starts the run and reviews the comparison; may download a report. Viewing an existing result never retrains.
Site adaptation	Dataset-specific label mapping, feature names and units, missing-value policy, classes and exclusions are documented. The dataset is de-identified and approved for the project.

2.2 Product functions

- Import the prepared tabular dataset and validate columns, types, ranges and target labels.
- Apply reproducible preprocessing fitted on training data only.
- Create one stratified train/validation/test split shared by all three model families.
- Train ML, DL and QML independently; record configuration, seeds and versions.
- Evaluate all models on the same untouched test records and compare them side by side.
- Display metrics, confusion matrices, threshold and limitations; optionally predict one record.
- Export a report with dataset version, split protocol, metric definitions and model configuration, without participant-level records.

2.3 User characteristics

Users are students or instructors with basic familiarity with tabular data and classification metrics. The site explains metrics and states that output is non-diagnostic. It is not intended for public or clinical use.

2.4 Constraints

- Dataset and label quality limit how far results generalize; NHANES is cross-sectional.
- QML may need feature reduction or a small feature subset because of simulator and circuit limits; the same reduced feature set is used for all three models in the main comparison.

- Split first, then fit imputers, scalers, encoders, feature selection and any resampling on training data only. Oversampling, if used, happens inside training folds; test prevalence stays natural.
- Submission deadline set by the instructor: 12 October 2026.

2.5 Assumptions and dependencies

- The prepared NHANES files can be legally used and the label is interpretable as documented in Section 4.1.
- Each class has enough records for a defensible split; otherwise, the limitation is reported with a validation alternative.
- The environment supports the chosen ML/DL libraries and a QML simulator.
- No, medically validated threshold is supplied; any threshold is experimental only.

2.6 Apportioning of requirements

Live quantum-processor use, clinical-system integration, prospective validation, population calibration, user accounts, public hosting and regulatory approval are out of scope.

3. Specific Requirements

3.1 External interface requirements

ID	Requirement	Priority	Verified by
UI-01	Every result view shall show the model name and dataset version.	Must	TC-07
UI-02	Every metric table shall label the evaluation split (test) and provide metric definitions.	Must	TC-07
UI-03	A non-diagnostic warning shall appear wherever an individual estimate is shown.	Must	TC-08
UI-04	A metric that cannot be computed shall be shown as “n/a”, never as 0.	Must	TC-07
HW-01	The website shall show whether each run used CPU, GPU or the QML simulator, and explain when a requested backend is unavailable.	Should	TC-05, TC-14
SW-01	The system shall reject unsupported file schemas with an error naming the problem. Model adapters shall accept the same feature matrix and return a positive-class probability or score.	Must	TC-02, TC-05
COM-01	Any remote API shall use authenticated HTTPS. Default processing is local; records and predictions shall not be sent to analytics or external services.	Must	TC-21

3.2 Use case diagram

Primary actor: **Student / Research User**. Secondary (external) actor: **NHANES public-use files**, which are the data source. All use cases lie inside the system boundary of the Early Metabolic Risk Detection Website.

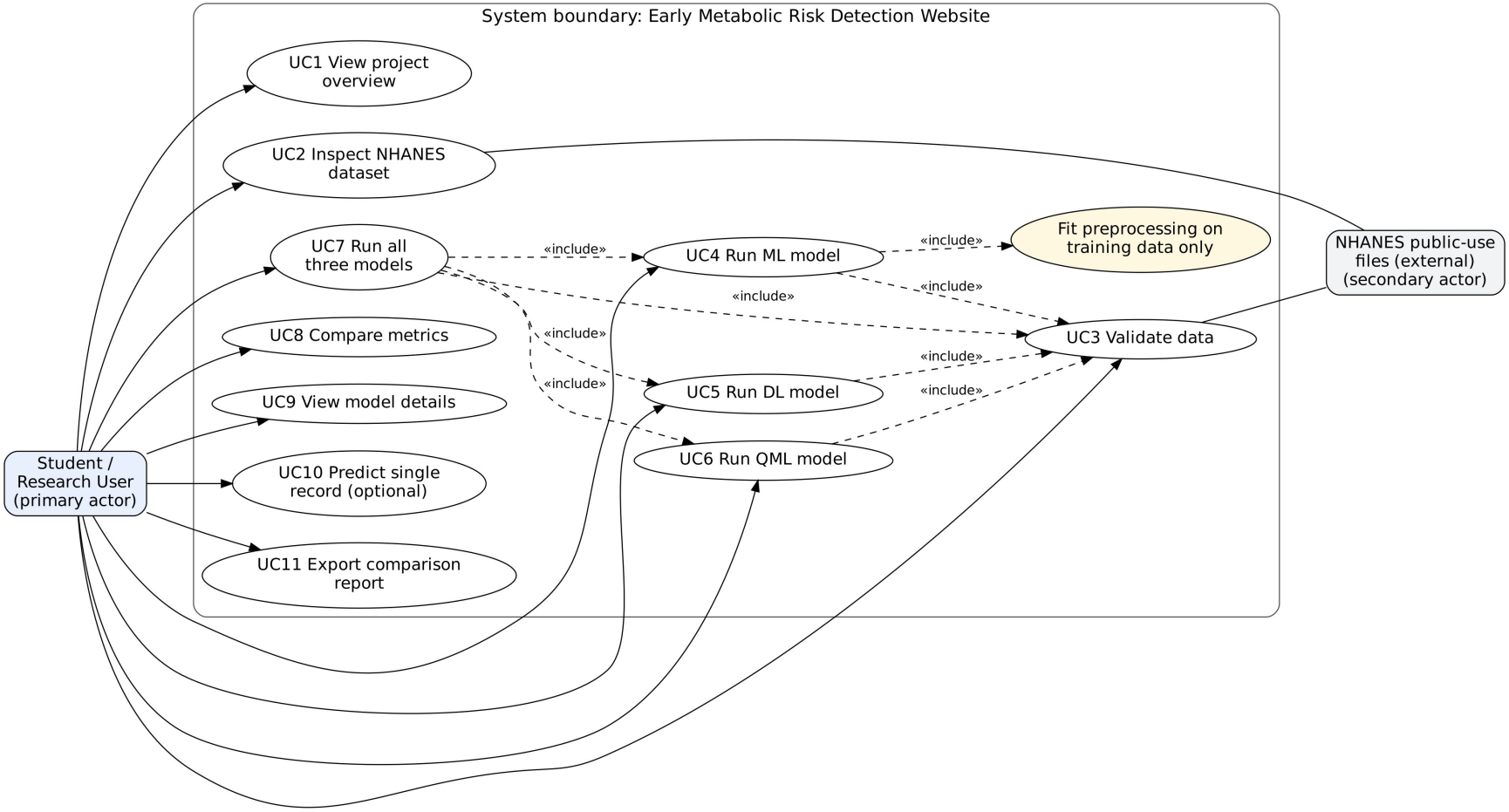


Figure 1: UML use case diagram – Early Metabolic Risk Detection Website

ID	Use case	Brief description	Linked FR
UC1	View project overview	User reads the purpose, dataset, model families and educational-use limitation.	FR-13
UC2	Inspect NHANES dataset	User sees cycle, label definition, eligible sample count, class balance and features.	FR-14
UC3	Validate data	System checks columns, types, ranges, missingness and target classes; blocks training on failure. Included by every run use case.	FR-02, FR-03
UC4-UC6	Run ML / DL / QML model	User starts one model; system applies training-only preprocessing, trains, evaluates and stores the run.	FR-04 to FR-06, FR-15
UC7	Run all three models	User starts ML, DL and QML together on one shared split; progress and status per model are shown.	FR-15
UC8	Compare metrics	User views the side-by-side table, charts and confusion matrices on the same test set.	FR-07, FR-08, FR-16
UC9	View model details	User opens configuration, seed, versions, status, runtime and limitations of one model.	FR-18
UC10	Predict single record (optional)	User enters feature values; system validates, scores and shows result with threshold and warning.	FR-09
UC11	Export comparison report	User downloads an aggregate report with provenance and no participant rows.	FR-10, FR-17

3.3 Functional requirements

Priority: **Must** = needed for acceptance; **Should** = expected but may be deferred if time is short.

ID	Requirement	Priority	Verified by
FR-01	The system shall import the prepared NHANES CSV (up to 50 MB) and display the record count and column count after import.	Must	TC-01
FR-02	The system shall validate required columns, data types, value ranges, missing-value rate and target classes before any training, and list every issue found.	Must	TC-02, TC-03
FR-03	If validation fails, the system shall block training and show an error that names each failing field; no model run shall start.	Must	TC-02, TC-03
FR-04	The system shall create one stratified train/validation/test split (default 70/15/15, fixed seed) and use identical row indices for ML, DL and QML. Test rows shall never be used for fitting or tuning.	Must	TC-04
FR-05	The system shall fit imputation, scaling, encoding and feature selection on training rows only, then apply the fitted transforms to validation and test rows.	Must	TC-04
FR-06	The system shall train ML, DL and QML models separately and record for each: algorithm, hyper-parameters, seed, library versions and run status.	Must	TC-05
FR-07	The system shall compute accuracy, precision, sensitivity, specificity, F1, ROC-AUC (when scores exist) and the confusion matrix for each model on the same test rows.	Must	TC-06
FR-08	The system shall show the three models side by side and name the primary ranking metric, which is fixed before test evaluation.	Must	TC-07
FR-09	The system shall predict only for records that pass validation and shall display model name, score, class label, threshold and the non-diagnostic warning with each prediction.	Must	TC-08
FR-10	The system shall export dataset ID/version, split seed, metrics, configuration, runtime and limitations in every report.	Must	TC-10, TC-12

ID	Requirement	Priority	Verified by
FR-11	The system shall store the seed of every run so that a repeat with the same seed, data and versions reproduces the split exactly and ML metrics exactly; DL and QML shall stay within the tolerance in NFR-03.	Should	TC-10
FR-12	The system shall reject or remove identity fields (e.g. SEQN, names, record IDs) from model features and exclude them from all reports (mandatory if identity fields exist in the data).	Should	TC-09
FR-13	The website shall open to a project page that explains the purpose, the selected dataset, the three model families and the educational-use limitation.	Must	TC-11
FR-14	The website shall let the user select the prepared NHANES dataset, validate its required columns, and show row counts and class counts before training.	Must	TC-01, TC-11
FR-15	The user shall be able to run ML, DL and QML separately or all three together; the website shall show progress and report a failed run with a readable message.	Must	TC-11, TC-14
FR-16	The comparison page shall show each model's accuracy, precision, sensitivity, specificity, F1 and ROC-AUC (when available) plus a confusion matrix, all on the same held-out records.	Must	TC-07, TC-12
FR-17	The user shall be able to download a comparison report with aggregate results and experiment settings and no participant-level records.	Must	TC-12
FR-18	The website shall let the user open a model-details view showing configuration, run status, runtime and known limitations.	Should	TC-12

3.4 Non-functional requirements

ID	Category	Requirement (measurable)	Verified by
NFR-01	Performance	For a file of up to 50,000 rows, validation feedback shall appear within 10 seconds on the documented reference laptop. Training runtime of each model shall be reported.	TC-13
NFR-02	Reliability	A failed run shall show an error and leave every earlier completed result unchanged (tested with at least 3 injected failures). A failed run shall never appear as current.	TC-14, TC-03
NFR-03	Reproducibility	With the same data, split seed and library versions: split indices and ML predictions shall be identical in 5 of 5 repeats. DL and QML shall be repeated with 5 seeds and reported as mean ± standard deviation.	TC-10
NFR-04	Security / privacy	Only approved de-identified data shall be used; identity fields shall be excluded; outputs shall be stored only in the team workspace (see Section 3.8).	TC-09, TC-21
NFR-05	Usability	A first-time student shall complete import, validation, run, comparison and export within 15 minutes using on-screen labels and the README only, without editing code (3 of 3 test users).	TC-15
NFR-06	Maintainability	Every model adapter shall implement the same interface (fit, predict_proba, describe). Replacing one adapter shall require no change to the other adapters, the evaluator or the UI.	TC-16
NFR-07	Portability	A clean machine shall run the website within 30 minutes by following the README and a single dependency file; library versions shall be recorded.	TC-17
NFR-08	Readability	All tables and plots shall carry axis labels and legends and shall not rely on colour alone; text contrast shall be at least 4.5:1; charts shall be readable in grayscale.	TC-18

ID	Category	Requirement (measurable)	Verified by
NFR-09	Responsiveness	On the reference computer the dashboard shall show a visible run status within 2 seconds of a run request and render a completed comparison within 5 seconds after metrics are ready.	TC-11, TC-19
NFR-10	Compatibility	The website shall work in the latest Chrome and Firefox at 1366×768 or larger, without horizontal scrolling.	TC-19
NFR-11	Resource logging	Runtime and peak memory shall be recorded for each model run. If a memory limit is exceeded the run shall stop with a message and not corrupt earlier results.	TC-22

3.5 Performance requirements

Predictive performance is an experimental outcome, not a guarantee. All three families are evaluated on identical held-out records with the same positive-class definition. Reported values: accuracy, precision, sensitivity, specificity, F1, ROC-AUC (when calculable), confusion matrix, threshold, class prevalence and confidence intervals where the sample allows. Tuning uses training data or cross-validation only; the test set stays untouched until final evaluation. The primary ranking metric is chosen before looking at test results; if finding elevated-risk people is the priority, **sensitivity at an agreed minimum specificity** is the primary metric and accuracy is secondary. Ties are broken by ROC-AUC. Runtime, memory and QML circuit/simulator settings are reported with the scores.

3.6 Design constraints

- The three model families remain separately implemented and identifiable.
- All three models use the same predictor columns, target definition, positive class, split and metric code. If QML needs fewer inputs, a smaller shared feature set is chosen before training and used for all three; any full-feature experiment is labelled separately.
- Preprocessing and feature selection are fitted within training data or folds only.
- QML dimensionality reduction, encoding and circuit configuration are documented.
- Output is never described as a diagnosis or medical advice.

3.7 Software system attributes

- **Reliability:** concise, actionable errors; a failed run never silently reuses stale metrics (NFR-02).
- **Availability:** continuous uptime is not required; local use works offline after dependencies are installed.
- **Maintainability:** data loading, preprocessing, model training, metrics and presentation are separate modules (NFR-06).
- **Fairness and limits:** subgroup performance is assessed where subgroup sizes allow and marked unstable when small; dataset limitations, label provenance, imbalance and exclusions are documented.

3.8 Security requirements

The project handles health-related survey data, so the website must protect confidentiality and integrity even though it runs locally. These are academic controls and do not claim clinical-grade security.

Security objectives

ID	Objective	Meaning
SO-1	Confidentiality	Prevent unauthorized disclosure of health-related records, features and predictions.
SO-2	Integrity and provenance	Preserve the integrity and traceability of datasets, splits and results so that a

ID	Objective	Meaning
		reported score can be reproduced and trusted.
SO-3	Safe input handling	Ensure uploaded files and user inputs cannot harm or take control of the server or the page.

Security requirements

ID	Requirement	Objective	Verified by
SR-1	Only approved de-identified or synthetic data shall be used, and identity columns shall be excluded from features.	SO-1	TC-09
SR-2	Stored data and results shall be restricted to the project team workspace; raw data and secrets shall be excluded from public repositories (via .gitignore).	SO-1	TC-21
SR-3	Every report shall record dataset version, split seed, model configuration, library versions and a timestamp.	SO-2	TC-10
SR-4	External data transfer shall be disabled by default; any remote service shall require team approval and authenticated HTTPS.	SO-1	TC-09, TC-21
SR-5	File upload shall accept only .csv files up to 50 MB; content shall be parsed as data only, never executed, and rendered as escaped text; other files shall be rejected with an error.	SO-3	TC-20
SR-6	The server shall listen on 127.0.0.1 by default, and no passwords, keys or tokens shall be stored in source code (use environment variables).	SO-1, SO-3	TC-21

4. Supporting Information

4.1 Dataset specification

The project uses the public NHANES 2017-2018 cycle. Selected files are joined on SEQN, and SEQN is then removed so it cannot act as a predictor. “Early metabolic risk” here means a prediabetes-range HbA1c result among adults in the survey. It is a practical academic label, not a claim that a model forecasts future disease.

Data element	Source / project use
Cycle	NHANES 2017-2018 public-use data.
Target measurement	GHB_J: LBXGH (glycohemoglobin, %).
Candidate predictors	DEMO_J: age and selected demographics; BMX_J: BMI and waist circumference; selected blood-pressure, lifestyle or health-history variables if documented.
Join key	SEQN is used only to merge files, then excluded from features and reports.
Binary label	Normal: LBXGH < 5.7%. Elevated risk: 5.7% ≤ LBXGH < 6.5%. Records with LBXGH ≥ 6.5% are omitted from this two-class experiment.
Eligibility / missing values	Adults aged 18 or older. Missing-data rules and final sample counts are documented after merging.
Leakage safeguards	LBXGH and any direct proxy are never inputs. Self-reported diabetes diagnosis, treatment and medication fields are excluded from predictors.
Evaluation note	NHANES is cross-sectional: results describe classification within the cycle, not prediction of later disease. Survey weights are recorded and the report states whether metrics are weighted.

4.2 Website pages and user flow

Page / area	What the user can do
Home / project overview	Read the purpose, dataset description, ML/DL/QML model families and the educational-use warning.
Dataset page	See NHANES cycle, label definition, eligible sample count, class balance, selected features and validation issues before training.
Run models	Choose ML, DL, QML or all three; start the run; see progress, completion, runtime or a readable error.
Comparison dashboard	Compare held-out results in a labelled table and charts; inspect confusion matrices and metric definitions.
Model details	Review algorithm or circuit, configuration, seed, software versions, run status, runtime and limitations.
Export	Download a report of aggregate metrics and provenance; no row-level participant data.

The comparison view puts the three models side by side and shows the ranking metric, class prevalence and confusion matrix so that a high accuracy cannot hide poor detection of the elevated-risk group.

Part B – Software Architecture and Design Specification

Prepared in the style of IEEE Std 1016-2009. Technology choices marked “proposed” can be changed by the team without changing the requirements.

1. Introduction

1.1 Purpose and scope

This part describes how the website in Part A will be built: its architecture, components, security design, interactions, API, data, and error handling. It traces every major requirement to a design element, so that the code and the test cases in Part C can be checked against Part A.

1.2 Design goals

- **Fair comparison:** one shared data pipeline, split and evaluator for ML, DL and QML.
- **Independence:** each model family is a replaceable adapter behind one interface.
- **Reproducibility:** seeds, versions and configuration are stored with each run.
- **Privacy:** local processing, no identity fields, aggregate-only reports.
- **Simplicity:** runs on one laptop for the course demonstration.

2. Architectural Design

2.1 Architecture pattern

The system uses a **layered client–server architecture** (presentation, application, model and data layers) with the **Adapter (Strategy) pattern** for the three model families and a **pipeline** for preprocessing and evaluation.

Pattern	Where it is used	Why
Layered client–server	Browser UI ↔ local web server ↔ model layer ↔ local data store	Separates presentation, logic and data; each layer can be tested alone (NFR-06).
Adapter / Strategy	MLAdapter, DLAdapter, QMLAdapter behind one ModelAdapter interface	Lets the orchestrator treat all models the same, so the comparison is fair and a model can be replaced without other changes (FR-06, NFR-06).
Pipeline	Validate → split → fit transforms on train → train → evaluate → report	Makes the order of operations explicit and prevents leakage (FR-04, FR-05).
Repository	Run Log and Results Store	One place for run records, written atomically so failures keep earlier results (NFR-02).

Alternatives considered: a single notebook (no UI, weak separation) and micro-services (extra deployment effort with no benefit for a local demo). Both were rejected.

2.2 Component diagram (Figure 2)

Figure 2 shows the main components and their dependencies. Component responsibilities follow in Section 2.3.

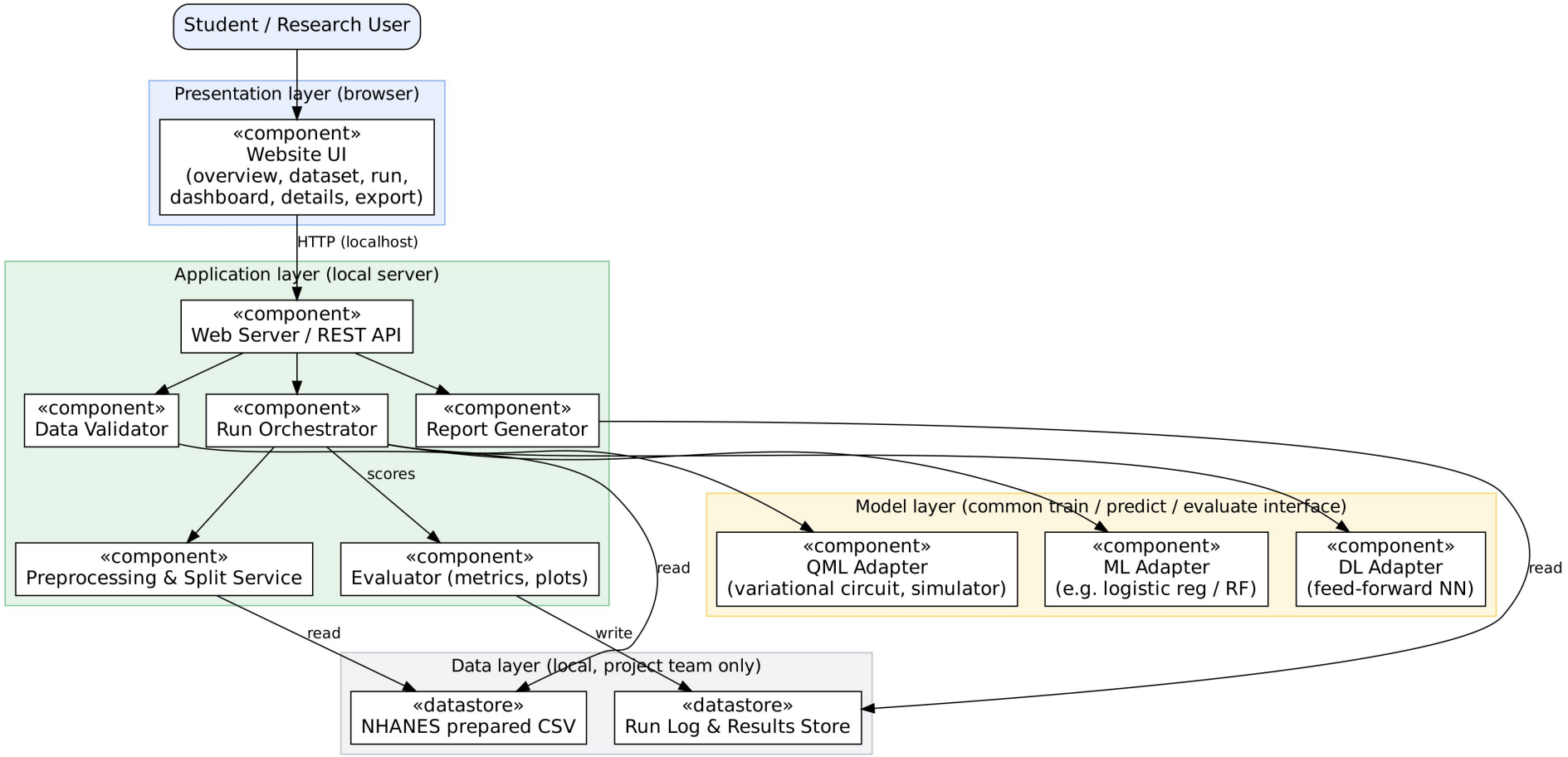


Figure 2: UML component diagram – layers and components

2.3 Component descriptions

Component	Responsibility	Interfaces
Website UI	Home, dataset, run, comparison dashboard, model details and export pages; shows warnings, progress and errors.	Calls the REST API over HTTP.
Web Server / REST API	Receives requests, checks inputs, starts runs, returns status, results and reports in JSON; maps errors to the error format in Section 3.6.	Exposes the API in Section 3.5.
Data Validator	Checks schema, types, ranges, missingness, target classes, identity fields and file type/size.	validate(data) → ValidationReport
Preprocessing & Split Service	Creates the shared stratified split; fits imputers/scalers/encoders on training rows only; applies them to validation and test rows.	make_split(), fit_transform_train(), transform()
Run Orchestrator	Runs the pipeline for the selected models, tracks status, isolates failures between models, stores run records.	start_run(), get_status()
ML / DL / QML Adapters	Wrap a classical classifier, a feed-forward neural network and a variational quantum circuit (simulator); return probabilities and metadata.	ModelAdapter: fit, predict_proba, describe
Evaluator	Computes the common metrics and confusion matrices on identical test rows; reports “n/a” for unavailable metrics.	metrics(), confusion()
Report Generator	Builds the aggregate-only report with provenance.	build(run_ids)
Run Log & Results Store	Local files or database holding run records, seeds, versions and metrics.	Read/write by Orchestrator, Evaluator, Report Generator

2.4 Proposed technology

- Language and server: Python 3.11 with a lightweight web framework (Flask or FastAPI).
- ML: scikit-learn. DL: PyTorch (CPU, GPU optional). QML: PennyLane or Qiskit on a simulator.
- Front end: HTML, CSS and JavaScript with a charting library; labelled charts with non-colour cues (NFR-08).
- Storage: local files (CSV/JSON) or SQLite inside the team workspace; versions pinned in one requirements file (NFR-07).

2.5 Deployment view

Everything runs on one laptop: the browser connects to the local server on 127.0.0.1; the server calls the model adapters in the same process or in worker processes; the QML adapter uses a local simulator. No cloud service is required.

2.6 Traceability of requirements to design

Requirement(s)	Component(s)	Design element
FR-01, FR-02, FR-03, FR-14	Website UI, Web API, Data Validator	POST /api/datasets/validate; ValidationReport
FR-04, FR-05	Preprocessing & Split Service, Run Orchestrator	Shared SplitIndices; transforms fitted on train rows only
FR-06, NFR-06	ML / DL / QML Adapters, RunRecord	ModelAdapter interface; RunRecord stores config, seed, versions
FR-07, FR-08, FR-16, UI-01 to UI-04	Evaluator, Website UI dashboard	GET /api/comparison; “n/a” for missing metrics

Requirement(s)	Component(s)	Design element
FR-09, UI-03	Web API, saved pipeline, Website UI	POST /api/predict; warning attached by server
FR-10, FR-11, FR-17, SR-3	Report Generator, Run Log	GET /api/reports/{runId}; seed and versions stored
FR-12, SR-1, NFR-04	Data Validator, Preprocessing	Identity-field blocklist (SEQN, name, id)
FR-13, FR-15, FR-18	Website UI, Run Orchestrator	POST /api/runs; GET /api/runs/{id}; GET /api/models/{family}
NFR-01	Data Validator	Vectorised column checks
NFR-02, NFR-11	Run Orchestrator, Run Log	Atomic write of completed runs; per-model failure isolation; memory/time logging
NFR-03	Orchestrator, Adapters	Seed passed to split and every adapter
NFR-05, NFR-08, NFR-09, NFR-10	Website UI	Labelled layout, status polling, supported browsers
NFR-07	Packaging	requirements file and README
SR-2, SR-4, SR-5, SR-6	Web API, deployment config	See security architecture (Section 2.7)

2.7 Security architecture

Trust boundaries. (1) Browser to server: all input is untrusted. (2) Server to local storage: only the team workspace is trusted. (3) Server to the outside network: closed by default.

Threat	Control in the design	Requirement	Test
Disclosure of participant data through export or logs	Identity fields removed at validation; reports built from aggregate metrics only; row-level data never logged.	SR-1, FR-12, FR-17	TC-09
Unintended data leaving the machine	Server bound to 127.0.0.1; no analytics or third-party calls; remote use only with HTTPS and authentication after approval.	SR-4, SR-6	TC-21
Malicious or malformed upload	Extension and size check (.csv, ≤ 50 MB); parsed as data only; cells never evaluated; output HTML-escaped.	SR-5	TC-20
Raw data or secrets leaked via the repository	.gitignore excludes raw data and credentials; secrets in environment variables.	SR-2, SR-6	TC-21
Wrong or stale results trusted as current	Every report and result carries dataset version, seed, configuration, versions and timestamp; failed runs never replace earlier results.	SR-3, NFR-02	TC-10, TC-14
Output mistaken for medical advice	Non-diagnostic warning attached by the server to every estimate and shown on all result pages.	UI-03	TC-08

3. Detailed Design

3.1 Sequence diagram: train and compare three models

Figure 3 covers use cases UC3 and UC7 including the validation-failure path (FR-02, FR-03, FR-04 to FR-08, FR-15).

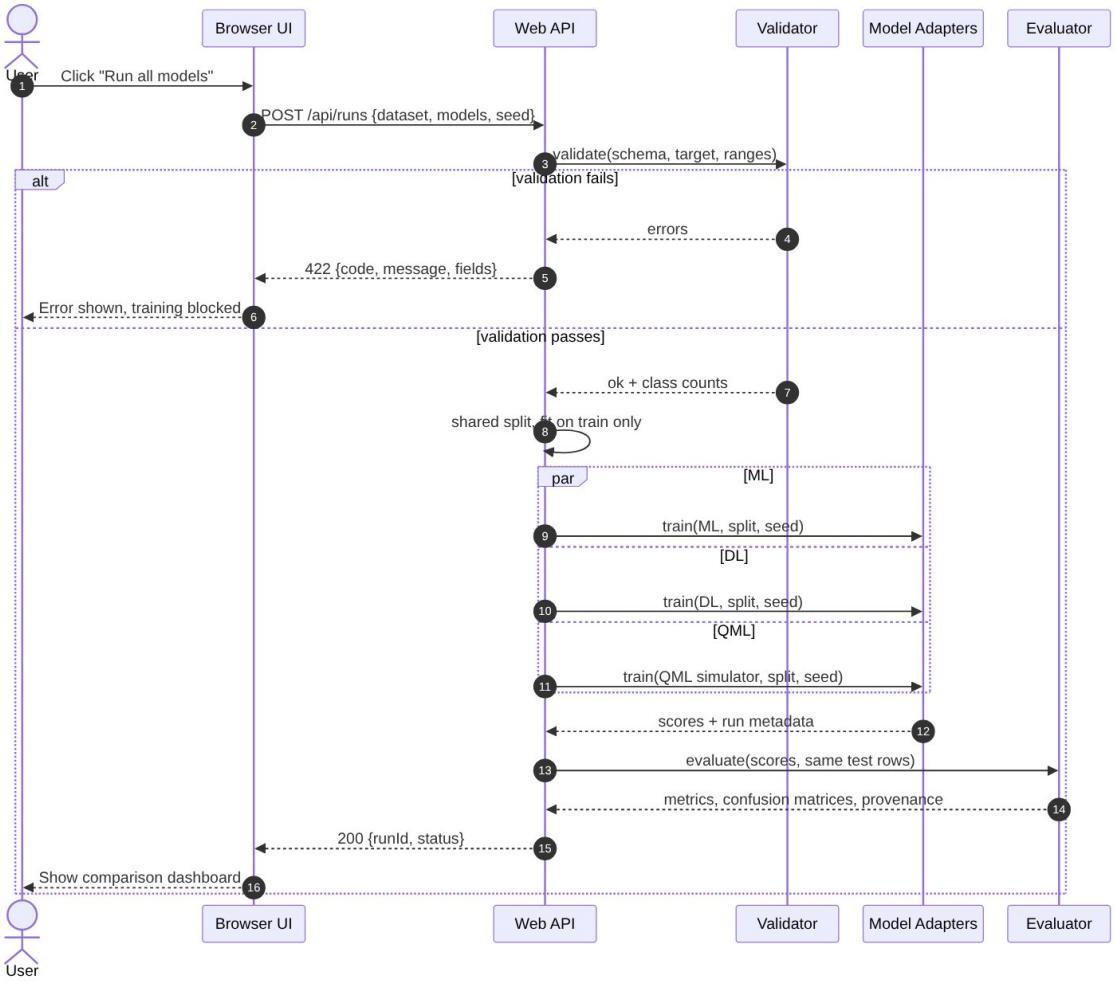


Figure 3: Sequence diagram – validate, train ML/DL/QML and compare

3.2 Sequence diagram: individual estimate

Figure 4 covers UC10, the optional single-record estimate (FR-09, UI-03).

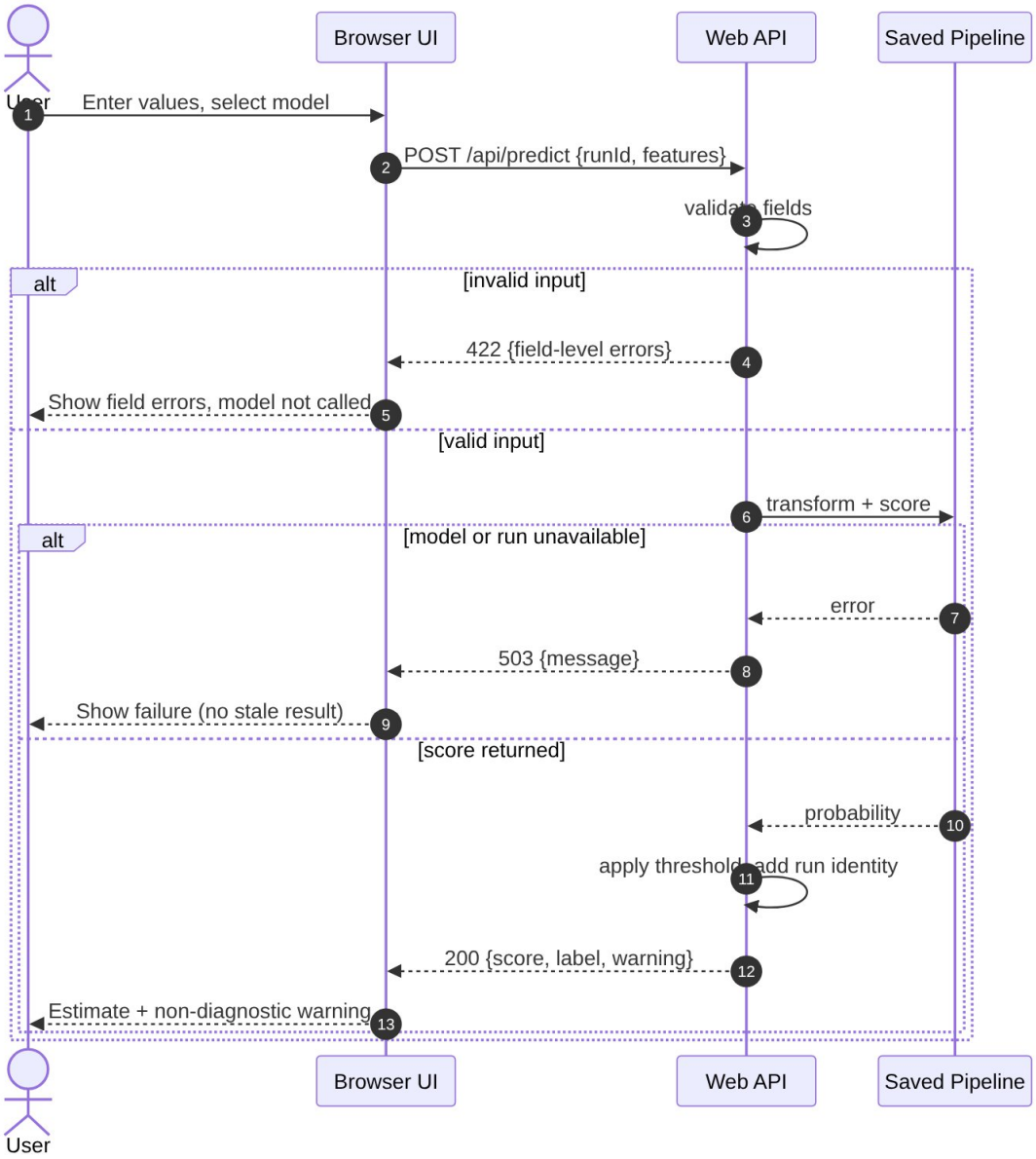


Figure 4: Sequence diagram – individual estimate with validation and failure paths

3.3 Sequence diagram: export report

Figure 5 covers UC11 (FR-10, FR-17, SR-3).

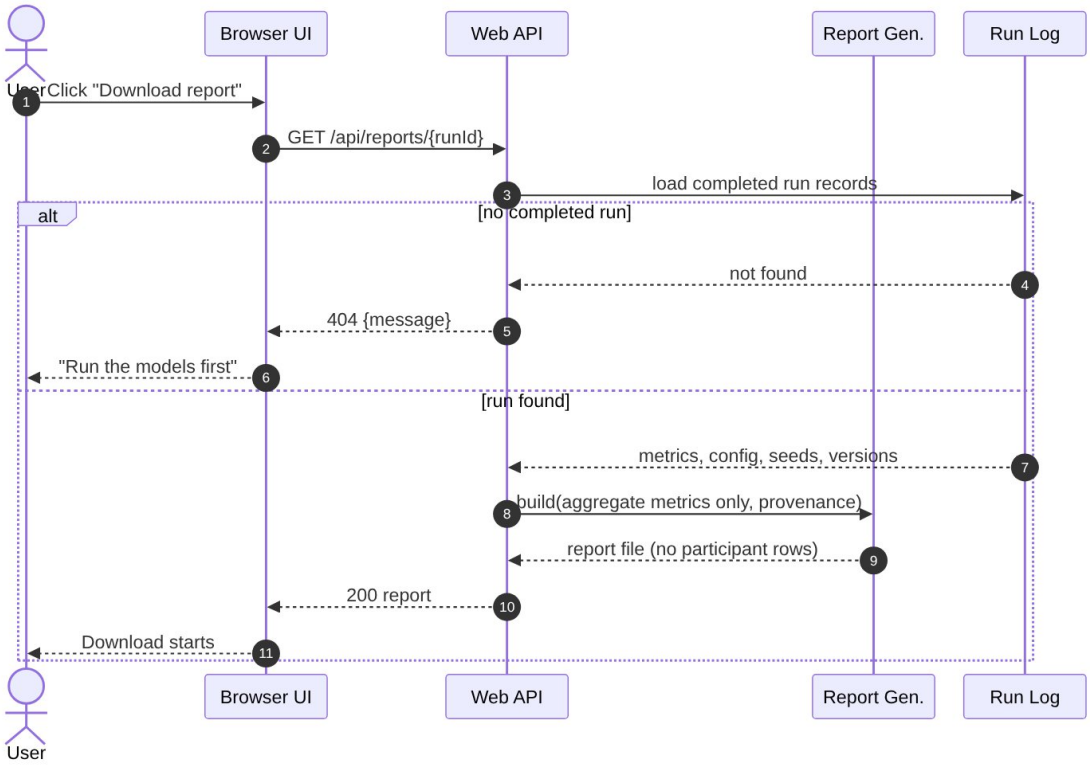


Figure 5: Sequence diagram – export aggregate report

3.4 Class design (Figure 6)

The key classes and the ModelAdapter interface that makes the three model families interchangeable (NFR-06).

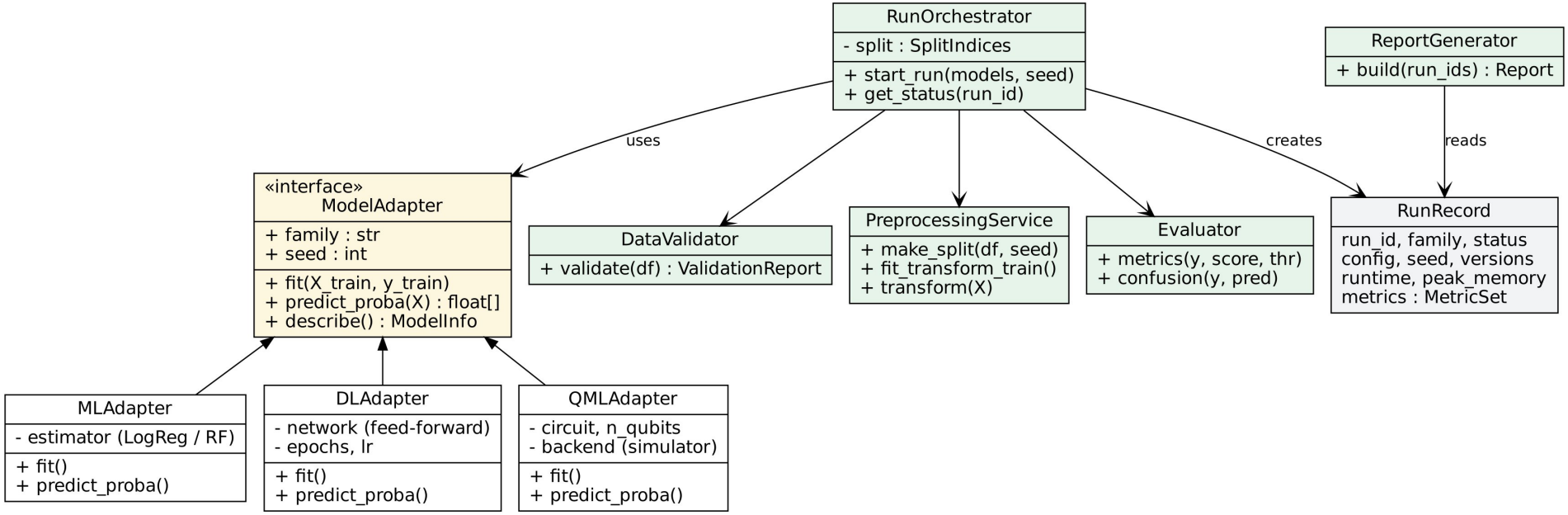


Figure 6: UML class diagram – adapters, orchestration and services

3.5 API design

REST over HTTP on localhost; JSON bodies; all times in ISO 8601. All endpoints are prefixed with /api.

Method	Endpoint	Purpose	Request → success response / errors
GET	/dataset	Dataset information (cycle, label definition, counts, features).	→ 200 {cycle, label, rows, classCounts, features[]}. 404 DATASET_NOT_PREPARED.
POST	/datasets/validate	Validate the prepared dataset or an uploaded CSV.	{datasetId} or multipart .csv → 200 {valid, rows, columns, classCounts, issues[]}. 413 FILE_TOO_LARGE, 415 FILE_TYPE_UNSUPPORTED, 422 SCHEMA_INVALID.
POST	/runs	Start a run for one, two or three model families.	{datasetId, models[“ml”,“dl”,“qml”], seed} → 202 {runId, status}. 422 VALIDATION_FAILED, 409 RUN_IN_PROGRESS.
GET	/runs/{runId}	Progress and status per model.	→ 200 {runId, models[{family, status, progress, runtimeSec, error?}]}. 404 RUN_NOT_FOUND.
GET	/comparison?runId=	Side-by-side metrics on the test set.	→ 200 {testSet{n, prevalence}, rankingMetric, models[{family, accuracy, precision, sensitivity, specificity, f1, rocAuc|null, confusion}]}. 404 RUN_NOT_FOUND, 409 RUN_NOT_COMPLETE.
GET	/models/{family}?runId=	Model details view.	→ 200 {family, config, seed, versions, status, runtimeSec, peakMemoryMB, limitations}. 404 RUN_NOT_FOUND.
POST	/predict	Optional single-record estimate.	{runId, family, features{}} → 200 {score, label, threshold, model, runId, warning}. 422 FIELD_INVALID, 503 PREDICTION_UNAVAILABLE.
GET	/reports/{runId}	Download the aggregate report.	→ 200 file (JSON/PDF/CSV). 404 RUN_NOT_FOUND.

Example request and response (start run)

```
POST /api/runs
{ "datasetId": "nhanes-2017-2018-v1", "models": ["ml","dl","qml"], "seed": 42 }

202 Accepted
{ "runId": "r-0007", "status": "queued" }
```

Example response (comparison, abbreviated)

```
200 OK
{ "testSet": { "n": 1200, "prevalence": 0.31 },
  "rankingMetric": "sensitivity@spec>=0.70",
  "models": [ { "family": "ml", "sensitivity": 0.0, "rocAuc": null,
                "confusion": [[0,0],[0,0]] } ] }
// values above are placeholders, not results
```

3.6 Error handling

Principles: (1) every error returns the same JSON shape; (2) messages are actionable and never show stack traces; (3) a failure in one model does not stop the others; (4) a failed run never replaces or relabels an earlier completed result; (5) every error is logged with a requestId; (6) unavailable metrics are shown as “n/a”.

```
{ "error": { "code": "SCHEMA_INVALID",
             "message": "Target column is missing.",
             "fields": ["label"], "requestId": "q-4821" } }
```

Error code	HTTP	Situation	System behaviour
FILE_TYPE_UNSUPPORTED	415	Upload is not a .csv file.	Reject; nothing is parsed or stored.
FILE_TOO_LARGE	413	Upload exceeds 50 MB.	Reject with size limit in message.
SCHEMA_INVALID / VALIDATION_FAILED	422	Missing column or target, wrong type, out-of-range values.	List every failing field; block training.
FIELD_INVALID	422	Single-record input fails validation.	Field-level errors; model is not called.
RUN_IN_PROGRESS	409	A run is already active.	Refuse new run; show current status.
RUN_NOT_FOUND	404	Unknown runId.	Message asks the user to run the models first.
RUN_NOT_COMPLETE	409	Comparison requested before completion.	Show progress, not partial metrics as final.
MODEL_FAILED	200 (failed status)	One adapter raises an error.	Mark that model failed with reason; other models continue; earlier results unchanged.
BACKEND_UNAVAILABLE	503	GPU or QML backend missing.	Fall back to CPU/simulator if allowed, and say so.
OUT_OF_MEMORY	200 (failed status)	Memory limit exceeded.	Stop the run gracefully; record peak memory; keep earlier results.
PREDICTION_UNAVAILABLE	503	Saved model or run not available.	Show failure; never show a previous result as current.

3.7 Data design

Entity	Key fields
Dataset	datasetId, cycle, version, source files, download date, label definition, row count, class counts, feature list.
SplitIndices	splitId, seed, train/validation/test row indices, stratification column.
RunRecord	runId, family, status, configuration, seed, library versions, backend, runtime, peak memory, timestamp, error.
MetricSet	accuracy, precision, sensitivity, specificity, F1, ROC-AUC or null, confusion matrix, threshold, test prevalence.
Report	runIds, dataset version, split seed, metric table, configuration, runtime, limitations. No participant rows.

3.8 User interface design

The pages are those listed in Part A Section 4.2. Design rules: (1) the non-diagnostic warning and dataset/model identity appear on every result page; (2) status messages appear within 2 seconds of a run request; (3) charts have labels, legends and patterns, not colour alone; (4) the layout works at 1366×768 without horizontal scrolling.

Part C – Software Test Plan and Test Cases

Prepared in the style of IEEE Std 829-2008 (test plan). Section 5.1 holds the security validation; Section 11 holds the test cases, and Section 12 the traceability to the SRS.

1. Introduction

1.1 Test plan identifier

Identifier	STP-CODECRAFT-EMRD-1.0
Project	Early Metabolic Risk Detection Website using ML, DL and QML
Version / date	1.0 / 9 October 2026 (submission due 12 October 2026)
Related documents	Part A (SRS) and Part B (Architecture and Design) of this document

1.2 Purpose and scope

This plan describes how the team will check that the website works from data selection to report export, that private information is handled carefully, and that the ML, DL and QML models receive a fair comparison. Clinical validation is outside the project.

1.3 References

- Part A – SRS and Part B – Design of this document.
- IEEE Std 829-2008; IEEE Std 830-1998; IEEE Std 1016-2009.
- NHANES 2017-2018 documentation (CDC/NCHS).

2. Test Items

The following items of the v1.0 prototype are under test:

Test item	Description
Data Validator	Schema, type, range, missingness, target, identity-field and file checks.
Preprocessing & Split Service	Shared stratified split and training-only transforms.
ML, DL and QML Adapters	Model training and probability output behind the common interface.
Evaluator	Metric and confusion-matrix computation on identical test rows.
Web Server / REST API	Endpoints, input checking, error responses.
Website UI	Pages, navigation, progress display, dashboard, warnings.
Report Generator	Aggregate-only report with provenance.

3. Features to be Tested

Feature	Requirements	Test cases
Data import and validation	FR-01, FR-02, FR-03, FR-14, SW-01	TC-01 to TC-03
Shared split and leakage control	FR-04, FR-05	TC-04
Separate model training and metadata	FR-06, FR-11, HW-01	TC-05, TC-10

Feature	Requirements	Test cases
Metrics and comparison	FR-07, FR-08, FR-16, UI-01 to UI-04	TC-06, TC-07
Individual estimate	FR-09, UI-03	TC-08
Website workflow, details and export	FR-13, FR-15, FR-17, FR-18, FR-10	TC-11, TC-12
Privacy and security	FR-12, NFR-04, SR-1 to SR-6, COM-01	TC-09, TC-20, TC-21
Performance, reliability, reproducibility	NFR-01, NFR-02, NFR-03, NFR-09, NFR-11	TC-10, TC-13, TC-14, TC-19, TC-22
Usability, maintainability, portability, readability, compatibility	NFR-05 to NFR-08, NFR-10	TC-15 to TC-19

4. Features Not to be Tested

- Clinical validity of the HbA1c-based label or any claim of diagnostic accuracy.
- Use of a real quantum processor, clinical-system integration, user accounts and public hosting (out of scope, Part A Section 2.6).
- Load above 50,000 rows and concurrent multi-user use.
- Third-party library internals (scikit-learn, PyTorch, the QML framework); only their use is tested.

5. Approach

5.1 Security validation

Security validation checks the objectives SO-1 to SO-3 and requirements SR-1 to SR-6 of the SRS. Test cases: TC-09, TC-20 and TC-21.

Check	Pass condition	Method	Requirement / test
Approved data	Source, licence and de-identification recorded; synthetic fixture has no real identifiers.	Document review	SR-1 / TC-09
Feature privacy	Identity fields rejected or removed; report has no row-level personal data.	Inject identifier column; inspect export	SR-1, FR-12 / TC-09
Local handling	No external transfer by default; remote service only with approved authenticated HTTPS.	Network capture; config review	SR-4, COM-01 / TC-09, TC-21
Result integrity	Report records dataset version, split seed, configuration, versions and timestamp.	Report inspection after repeat runs	SR-3 / TC-10
Access control	Files stay in the approved team workspace; repository excludes raw data and secrets.	Repository and .gitignore review	SR-2, SR-6 / TC-21
Upload safety	Only .csv up to 50 MB accepted; content never executed or rendered as HTML.	Malicious-file tests	SR-5 / TC-20
Listening address	Server listens on 127.0.0.1 by default.	Port/address check	SR-6 / TC-21

5.2 Test levels and types

- **Unit:** validators, preprocessing, metric functions, threshold logic, report serialization.
- **Integration:** dataset-to-model pipeline for each family and the shared evaluator; API to UI.
- **System:** full workflow from import through comparison and export, in the browser.

- **Security and privacy:** identity exclusion, approved data, local-only behaviour, upload safety.
- **Non-functional:** performance, reliability, reproducibility, usability, portability, readability, compatibility.
- **Model evaluation:** common split and metrics, training-only tuning, untouched test evaluation, resource logging.
- **Manual:** warning text, metric labels, plot readability, reproducibility steps.

5.3 Model comparison protocol

- Define the positive class and label mapping before training.
- Create one stratified split (or one documented repeated-CV protocol) and reuse the exact indices for all models.
- Fit imputation, encoding, scaling, feature selection and resampling only on training data or folds.
- Tune on training/validation data only; never choose a threshold from test data.
- Evaluate each final model once on identical untouched test rows; report metrics, threshold, prevalence, confusion matrix and uncertainty where feasible.
- Choose the primary metric before seeing test results (sensitivity at an agreed minimum specificity; accuracy secondary).
- Never claim that the highest score proves clinical superiority; discuss sample size, distribution shift, subgroup limits and QML resource cost.

5.4 Test data and environment

Record OS, CPU/RAM, optional GPU, QML simulator, language and library versions of the reference computer. Use a small synthetic fixture for deterministic tests and the approved de-identified NHANES data for end-to-end evaluation. Never test with identifiable patient data. Record dataset source, licence, label mapping, feature units and prevalence.

6. Item Pass/Fail Criteria

A test case **passes** when the actual result matches the expected result in Section 11. Defects are rated:

Severity	Meaning
Critical	Privacy breach, data leakage into the test set, wrong metrics, or a failed run shown as current.
Major	A Must requirement fails but a workaround exists.
Minor	Cosmetic or wording problem that does not change results.

7. Entry, Exit, Suspension and Resumption Criteria

Entry criteria	Exit criteria
Requirements and target label documented; synthetic test fixture available; model interfaces implemented; library versions recorded.	All Must-priority test cases pass; no open Critical defect; all models use the shared split and evaluation; metrics and limitations recorded; every failed test explained.

Suspension: testing is paused if a data-leakage or privacy defect is found, the dataset becomes unavailable, or the test environment is broken. **Resumption:** after the fix, re-run the affected test cases and a smoke test (TC-01, TC-05, TC-07).

8. Test Deliverables

This test plan, completed execution record (Section 11.2), defect notes, model configurations, run logs, dataset description, split seed, metric table, confusion matrices, resource observations and the final comparison report.

9. Responsibilities and Schedule

Role	Responsibility
All members (Abhay TS, Aarav Yuval, Abhishek PH)	Write unit tests for their own modules; peer-test modules written by another member; log defects; run the final end-to-end tests together.
Instructor (Cvsn Reddy)	Reviews the plan and the results.

Phase	Timing
Test design (this document)	Complete for submission on 12 October 2026.
Unit tests	Alongside development of each module.
Integration and system tests	After all three adapters and the dashboard work.
Final model evaluation and security checks	After tuning is frozen; before the final demonstration.

10. Risks and Contingencies

Risk	Contingency
QML simulator is too slow for the full feature set	Use a smaller shared feature set (few qubits) for all three models; label any full-feature run separately.
Class imbalance hides poor detection of the elevated-risk group	Report sensitivity, specificity and confusion matrix, not accuracy alone; resample only inside training folds.
Data leakage through preprocessing order	Split first; leakage audit (TC-04) is a Critical test.
Missing values or small subgroups in NHANES	Documented imputation policy; mark small subgroup results as unstable.
Limited time before the deadline	Run all Must test cases first; Should-priority cases next.

11. Test Cases

11.1 Test case specification

There are 22 test cases: 11 functional (F), 8 non-functional (NF) and 3 security (S). The brief asks for at least 7 to 10; the extra cases cover every non-functional and security requirement. Replace the generic fixture values with the real NHANES values before execution.

ID	Test	Type	Input / action	Expected result	Traces to
TC-01	Valid import	F	Open Dataset page; select the prepared NHANES CSV that has the required features and target.	Import succeeds; record count, column count, class counts and dataset identifier are shown.	FR-01, FR-14
TC-02	Missing target	F	Import a CSV without the target/label column.	Validation fails with a message naming the missing target; no model starts.	FR-02, FR-03
TC-03	Invalid values	F	Fixture with a wrong data type, an out-of-range value and missing values.	Every issue is listed with its field name; documented policy is applied or the run is blocked; no crash.	FR-02, FR-03, NFR-02
TC-04	Shared split / leakage	F	Run all three models with a fixed seed; export split indices; inspect where imputer and scaler are fitted.	Identical train/validation/test indices for all models; transforms fitted on training rows only; no test row used in fitting or tuning.	FR-04, FR-05
TC-05	Separate training	F	Train ML, DL and QML on the fixture and on NHANES data.	Three run records, each with family, configuration, seed, library versions, backend and status.	FR-06, HW-01, SW-01
TC-06	Metric correctness	F	Evaluate a fixture with known predictions containing both classes.	Accuracy, precision, sensitivity, specificity, F1, ROC-AUC and confusion matrix equal an independent calculation to 4 decimals.	FR-07
TC-07	Comparison display	F	Open results for all three model families.	Same test set and positive class; ranking metric named; model name and dataset version shown; unavailable metric shown as “n/a”, not 0; metric definitions present.	FR-08, FR-16, UI-01, UI-02, UI-04
TC-08	Prediction safety	F	Submit one valid and one invalid single-record input.	Valid: model, score, class, threshold and non-diagnostic warning shown. Invalid: field-level error and the model is not called.	FR-09, UI-03
TC-09	Privacy checks	S	Add an identifier-like column (SEQN, name) to the CSV; inspect features, export and network traffic.	Identifier removed or rejected; export contains aggregate data only; no external request in the network log.	FR-12, NFR-04, SR-1, SR-4
TC-10	Export and reproducibility	F	Repeat a run 5 times with the same seed and configuration; export a report each time.	Split identical; ML metrics identical; DL/QML shown as mean ± SD; each report has dataset version, seed, configuration,	FR-10, FR-11, NFR-03, SR-3

ID	Test	Type	Input / action	Expected result	Traces to
				library versions, timestamp, metrics, runtime and limitations.	
TC-11	Website run workflow	F	Open the site in Chrome; view the home page; select NHANES; run one model, then all three.	Home page shows purpose and limitation; validation and class counts appear; progress is visible; each model shows a status; a clear message appears if a run fails.	FR-13, FR-14, FR-15, NFR-09
TC-12	Dashboard, details and export	F	Open completed results; compare metrics; open a model-details view; download the report.	Dashboard uses the same test set and shows confusion matrices; details show configuration, status, runtime and limitations; report has aggregate results and provenance, no participant rows.	FR-10, FR-16, FR-17, FR-18
TC-13	Validation performance	NF	Validate a 50,000-row synthetic file on the reference laptop and time it.	Validation feedback within 10 seconds; time recorded.	NFR-01
TC-14	Failure recovery	NF	Complete one run; start another and force the QML adapter to fail (e.g. disable the simulator).	Failed model is clearly labelled; other models' results are shown; the earlier completed run is unchanged; stale metrics are never shown as current.	NFR-02, FR-15
TC-15	Usability walkthrough	NF	Three first-time students follow the README: import, validate, run, compare, export, without editing code.	All three finish within 15 minutes without help; problems are logged.	NFR-05
TC-16	Maintainability	NF	Review the code; replace one adapter with a stub that implements the interface.	Application still runs; no change needed in other adapters, evaluator or UI; all adapters use fit / predict_proba / describe.	NFR-06
TC-17	Portability	NF	Follow the setup instructions on a clean machine or fresh virtual environment.	Website starts within 30 minutes using one dependency file; versions are recorded.	NFR-07
TC-18	Readability	NF	Review the dashboard and exported report; convert charts to grayscale; check labels and contrast.	All charts have axis labels, legends and non-colour cues; readable in grayscale; text contrast at least 4.5:1.	NFR-08
TC-19	Browser compatibility and responsiveness	NF	Open the site in the latest Chrome and Firefox at 1366×768; start a run; time the status and the dashboard.	No horizontal scrolling; run status within 2 s of the request; dashboard within 5 s after metrics are ready.	NFR-09, NFR-10
TC-20	Upload safety	S	Upload (a) an .exe renamed .txt, (b) a CSV over 50 MB, (c) a CSV whose cells contain formulas or script text (e.g. =cmd…, <script>).	(a) and (b) rejected with an error; (c) text is treated as data, never executed or rendered as HTML.	SR-5
TC-21	Network, access and secrets	S	Check the listening address; capture traffic during a full run; search the repository for keys and raw data; inspect .gitignore.	Server bound to 127.0.0.1; no analytics or external calls; no secrets or raw NHANES data in the repository; outputs only in the team workspace.	SR-2, SR-4, SR-6, COM-01
TC-22	Resource logging	NF	Run all models; then run the DL model with an artificially	Runtime and peak memory recorded per model; when the	NFR-11, NFR-02

ID	Test	Type	Input / action	Expected result	Traces to
			small memory limit.	limit is exceeded the run stops with a message and earlier results stay intact.	

11.2 Test execution record

Test ID	Status (Pass/Fail)	Evidence / run ID	Defect or notes
TC-01			
TC-02			
TC-03			
TC-04			
TC-05			
TC-06			
TC-07			
TC-08			
TC-09			
TC-10			
TC-11			
TC-12			
TC-13			
TC-14			
TC-15			
TC-16			
TC-17			
TC-18			
TC-19			
TC-20			
TC-21			
TC-22			

12. Traceability Matrix (SRS to Test Cases)

Requirement	Test case(s)	What is verified
FR-01	TC-01	Import and counts
FR-02, FR-03	TC-02, TC-03	Validation and blocking of invalid input
FR-04, FR-05	TC-04	Shared split and training-only transforms
FR-06	TC-05	Separate model runs with metadata
FR-07	TC-06	Metric correctness
FR-08, FR-16	TC-07, TC-12	Side-by-side comparison on the same test set
FR-09	TC-08	Prediction output and rejection of invalid input
FR-10, FR-17	TC-10, TC-12	Report content, aggregate-only export
FR-11	TC-10	Seed-based reproduction
FR-12	TC-09	Identity-field exclusion
FR-13	TC-11	Home page content
FR-14	TC-01, TC-11	Dataset selection, validation, counts
FR-15	TC-11, TC-14	Run controls, progress, failure message
FR-18	TC-12	Model-details view
UI-01, UI-02, UI-04	TC-07	Model/dataset labels, split label, “n/a” handling
UI-03	TC-08	Non-diagnostic warning
HW-01, SW-01	TC-05, TC-02	Backend shown; schema rejected
COM-01	TC-21	No external transfer
NFR-01	TC-13	Validation time ≤ 10 s for 50,000 rows
NFR-02	TC-14, TC-03, TC-22	Failure handling keeps earlier results
NFR-03	TC-10	Reproducibility
NFR-04	TC-09, TC-21	Privacy and storage
NFR-05	TC-15	Usability
NFR-06	TC-16	Common adapter interface
NFR-07	TC-17	Clean setup
NFR-08	TC-18	Readability
NFR-09	TC-11, TC-19	Response times
NFR-10	TC-19	Browser compatibility
NFR-11	TC-22	Runtime and memory logging
SR-1	TC-09	Approved data, identity columns excluded
SR-2	TC-21	Workspace-only storage, repository clean
SR-3	TC-10	Provenance in report
SR-4	TC-09, TC-21	External transfer disabled
SR-5	TC-20	Upload safety
SR-6	TC-21	Localhost binding, no secrets in code

13. Approvals

Name	Role	Signature	Date
Abhay TS (PES1UG24AM008)	Team member		
Aarav Yuval (PES1UG24AM004)	Team member		
Abhishek PH (PES1UG24AM013)	Team member		
Cvsn Reddy	Instructor		

Appendix A. Experiment Record Template

To be completed after the experiments are run. Leave results blank until real runs exist.

Field	Record after experiments
Dataset source / version / licence and target map	
Records, features, classes and prevalence	
Exclusions and missing-data policy	
Split protocol, seed and shared-index identifier	
Preprocessing and feature selection	
ML configuration and software version	
DL configuration and software version	
QML circuit / simulator and software version	
Held-out metrics, threshold and confusion matrix	
Runtime, peak memory and backend	
Limitations and subgroup analysis	
Primary metric and selection rationale	